



CYPRUS INTERNATIONAL UNIVERSITY

CMPE344

Database Management Systems and Programming II

Instructor: Prof Dr Melike Şah Direkoğlu

Project (15%)

Report Due Date: 17th of May 2026

Project Presentations: Lecture Hours in the week of 18-22 May 2026

In this project, you are asked to design and implement a database that will store all the necessary information for a specific domain. You can form groups up to 4 students. **Each group will be assigned to a project domain by the course coordinator. Please add your group member names to the provided Google spreadsheet on the Moodle Webpage.**

After designing the database, you will implement a software which will insert, delete and update data.

Your database design should contain at least **6 tables**. You are free to design the database anyway you like, but the following requirements should be considered:

Your database should have a user table that will allow users to login the system. This table will be used for authentication purposes. The system should also support creating user groups such as **admin, employee, customers, technical staff, etc.** Each group should have some specific role in the system. For example, employee prepare some items and have full access to the database. Administrators manage systems, and so on.

Design foreign keys to other tables, in order to, visualize and relate information from different tables.

Write queries to insert, delete or update data from tables.

Write PL/SQL blocks such as procedures, functions or triggers. Check the details below.

The project will have 5 phases:

DESIGN: The database should be designed and should be demonstrated with **ER Diagrams**. Foreign keys, primary keys, constraints, default values should also be indicated with relational model.

Use any cloud based database that you want **as Railway, Render, Supabase, or others to implement your tables.**

DEVELOP: **Write a software (GUI)** (with any preferred language such as JAVA, C#, PHP, Python, HTML/Javascript etc.) to insert/update/delete and show data from your database. You should develop at least 5 windows (screens) to manage 6 tables, for insert/update and delete record from the database.

QUERY AND RESULT DISPLAY FROM CLIENT SIDE USING THE DATABASE CONNECTION:

Write at least **5-7 SQL queries** retrieving, displaying and showing statistical information to help management (use **join operations, subqueries, group by, order by, sum, average, count, etc.**) and **at least 5 PL/SQL blocks** (procedures, functions or triggers)

DEPLOY: This year, you will use any cloud based platform that support database such as **Railway, Render, Supabase, or others to implement your tables. Create one project account in your preferred online platform and invite other project members as collaborator. In addition, you should create a simple user interface (GUI) that interacts with your database to insert, update, delete or view information from table records. Finally, deploy the project online as a web app or as a mobile app using the cloud server capabilities.**

GITHUB: All information about the project including ERD diagram, DDL, DML, SQL queries, GUI implementation codes must be published in a **GitHub page. Create one project repository in GitHub and invite other project members as collaborator. So that everyone has an access to it and can collaborate.**

At the end of this project, you will submit a report that should contain the following items **as well as present it in the class in the presentation week (18-22 May 2026):**

1. Introduction

- An introduction giving brief information and assumptions about the system
- Student number and names of all project members.

2. Database

- ER Diagram,
- Data Definition Language Details (DDL), including all necessary constraints,
- Data Manipulation Language (DML),
- Your SQL queries and 5 PL/SQL blocks and answers/outputs.

3. Software

- Provide your DB connection code

- Provide screenshots of all your working application interface as well as their codes and give some brief information about screen shot (both from the development environment and the user interface)

4. GitHub

Information about your GitHub repository (link to it, make sure it is public when you submit the report) and documentation details.

❖ Individual or group work (max 5 people) is accepted.